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(1) Research Program/Project/Study Title: Traditional Knowledge Systems on Upland Fishing of the Indigenous Peoples in Capiz, Philippines	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
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		Province:	Capiz
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☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		☐ Collaborative Research (within campus) ☒ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Fisheries		(8) Discipline/Field of Specialization Social Science
(9A) Abstract Traditional knowledge of fishing practices in the upland waters of Capiz is essential in understanding how traditional fishing gears and methods interplay between traditional knowledge systems, cultural resilience, and sustainable development. A phenomenological research design was employed that centers on the lived experiences and perspectives of indigenous people who possess invaluable insights into these traditional practices. The study extracted the essence of their experiences through in-depth interviews and bracketing preconceived notions, uncovering the multifaceted significance of traditional fishing practices. The data gathered were analyzed using "Hycner's Explication Process" which includes bracketing, delineating, clustering, summarizing, and extracting unique themes. The study revealed that traditional fishing gears and methods hold cultural significance for the Panay Bukidnon People and the Ati Community, encompassing sustainable fishing, resource conservation, and a community-based approach to conservation. Recommendations include a cultural preservation initiative, educational programs and materials to raise awareness, and conservation policies that respect traditional practices.		
(9B) Keywords <i>Traditional Knowledge, Indigenous Peoples, Fishing Gears</i>		
(10) Introduction Fishing communities, especially in less developed regions, possess diverse traditional and indigenous knowledge that mirrors long-standing beliefs and customs (FAO,2023). It is a collective body of knowledge and beliefs passed down through generations by oral transmission (Berkes, 1993) or as part of the learning experience with one another or the community. As such, traditional knowledge (TK) is part of a community's belief systems and collective values and preserves national identity as well as cultural diversity (Agillon, 2007). Globally, there is a growing acknowledgment of the significant contribution that indigenous communities' traditional knowledge can make to the modern management of natural resources (Sahadevan, 2016). These fisher's ecological knowledge can contribute to the sustainability and legitimacy of environmental planning and management (Bermudez et al., 2023). This knowledge resource has the potential to provide location-specific, finely		

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detailed insights into both spatial and temporal aspects, as well as management strategies and institutions (Sahadevan, 2016). However, it is seriously threatened due to the commercialization of fishing, the breakdown of traditional communal leadership and oral knowledge transmission systems, modern education, and the movement of younger generations to urban areas to work and/or study. Consequently, IFK, which has been orally transmitted for generations, has either been lost, not learned by the current generation or remains undocumented (Kitolelei et al., 2021).

In the Philippines, 62 indigenous cultural communities (ICCs) are spread across various islands (Macusi et al., 2023). Most of them live in the upland forests, retaining many of their attitudes, beliefs, practices, and way of life despite the influences of modern living. They consider the land and resources in their immediate community as their source of living and way of life (Agillon, 2007). However, due to acculturation, state and corporate intrusions into the ancestral domain have irreversibly altered the people's ways, and they are no longer isolated. The culture and tradition have been subjected to changes (Tawas et al., 1999). As a result, this traditional knowledge has a risk of loss in recent times (Kechu, 2023). Hence, it should be maintained and preserved as a seek of ecological sustainability and health perspective (Lagade et al., 2023).

In Capiz, many traditions and practices have long relied on its water treasures including the traditional fishing gears and methods as a foundation of sustenance, culture, and identity. These indigenous practices have come to represent not only the historical legacy of Capiz but also the key to its cultural resilience and sustainable development. Therefore, according to Biclár et al. (2018), the recording of knowledge, attitudes, and practices is significant in the preservation and conservation of the indigenous peoples' natural resources.

Thus, this research seeks to collect and document the traditional knowledge systems of the age-old fishing practices of Capiz by employing a phenomenological research design that embraces indigenous peoples lived experiences and perspectives in the upland waters. It seeks to unravel the intricate cultural traditions and ecological wisdom of these practices.

(11) Materials and Methods or Methodology

Research Design

A qualitative phenomenological approach was used in this study that primarily aimed to collect and document the traditional knowledge system of the Panay Bukidnon in Tapaz and Jamindan, and Ati communities in Dumarao on fishing gears and methods in the upland waters of Capiz. The qualitative method is used to understand people's beliefs, experiences, attitudes, behavior, and interactions (Phatak et al., 2013), and phenomenology is an approach to research that seeks to describe the essence of a phenomenon by exploring

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it from the perspective of those who have experienced it (Cambridge Online Dictionary, 2023).

Purposive sampling was used to gain an in-depth understanding and explore specific phenomena to understand the wholeness of the traditional knowledge possessed by the indigenous people through in-depth interviews.

Participants of the Study

Purposive sampling or non-probability sampling technique is used in selecting the participants of this study by intentionally selecting specific individuals based on predefined criteria and the researcher's judgment. The purposive sampling technique is the deliberate choice of a participant due to the qualities the participant possesses. It is a nonrandom technique that does not need underlying theories or a set number of participants (Bernard, 2002; Etikan et al., 2016). Informants were selected based on the age of the participants, who are *tumandok* (natives) in the said community based on the list of community elders given by the National Commission for the Indigenous Peoples (NCIP) in the Tapaz Cluster Office.

To address the ethical considerations in conducting this type of research, a permit from the NCIP (National Commission for the Indigenous Peoples) Region VI and informed consent were secured.

Explicitation of Data

The explicitation of data is done using Hycner's (1999) explicitation process. According to Hycner (1999), this process focuses more on investigating the constituents of an experience while keeping the context of the whole. The researchers convey the data as explicitly stated by the participant rather than analyzing or interpreting what the data intends to show (Groenewald, 2004; Medina et al., 2022). Moreover, the researchers acknowledge and reflect on their preconceived notions or biases about the topic by setting aside assumptions and prejudices throughout the process with an open and unbiased mindset. Hence, the researcher intentionally refrains from influencing "the participants understanding of the phenomenon" (Chan et al., 2013; Butler, 2016).

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Hycner's (1991) Explicitation Process

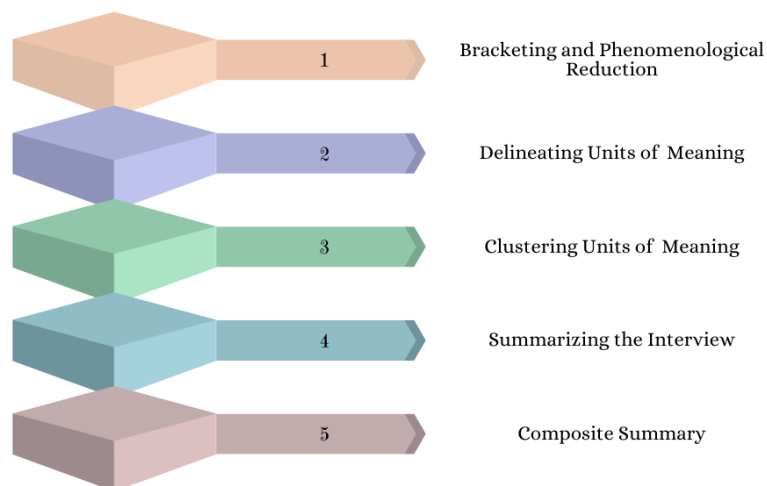


Figure1. Steps performed in data analysis using Hycner's (1991) Explicitation Process

(12) Results and Discussions

Below is the traditional knowledge possessed by the indigenous people on fishing gears and methods in the upland waters of Capiz collected through in-depth interviews.

1. Fishing Using Barehands called *Panikop* or *Igip*

The Panay Bukidnon of Jamindan, Capiz, captures small aquatic organisms through *panikop* or *igip*. *Panikop* is done simply by using bare hands and reversing the stone in the shallow portion of the river to catch *kagang* (Subphylum Crustacea). *Igip*, on the other hand, is similar to *panikop*. However, the difference is that it uses *moskitero* (fine mesh net) facing the water flow to filter and capture *unog* (family Gobiidae).

These fishing methods are still practiced then and now in catching small aquatic organisms, especially *kagang* and *unog*. Through this method, the abundance provided by the river can be sustained. Also, fish with *bihod* (fish roe) have less tendency to be in captivity and can still able to breed and produce more fish through this method.

2. Fishing Using *Sarapang* and *Sagangat*

Due to the difficulties brought by hand fishing, the indigenous people of Panay Bukidnon used a gear called *sarapang* and *sagangat* that imitates the function of the hand. *Sarapang* and *sagangat* are spears used to catch fish and crustaceans. It is a metal prong, usually a three-pronged spear (like a Poseidon trident), with a *sara-it* or *sima* (barbed points) on its rod that penetrates the prey's flesh. The prong is attached to the *kawayan* (bamboo)

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that serves as a handle for the gear. *Sarapang* is only specific for catching small aquatic organisms like *urang* (*Macrobrachium rosenbergii* sp.). At the same time, *sagangat* is much more prominent in size and is used in catching bigger fish like *sili* (*Anguilliformes*) and *karpa* (*Cyprinus carpio*).

Despite many fishing developments, the Panay Bukidnon still uses these gears even if it only catches a small amount of fish due to its cultural significance. They learned this traditional fishing knowledge from their *ginikanan* (parents) and continued to pass it down to the next generation.

3. Fishing Using *Bunit* or Pole and Line

Bunit or pole and line is a simple hand-operated fishing gear that can be operated in rivers and seas. The gear consists of a long pole made from *kawayan* and a nylon string with a modified hook made from umbrella wire placed at the end part of the line. The hook is baited to lure the fish.

In Dumarao, Capiz, the Ati first gathered an earthworm that would be used as a baiting element for the fishing operation. The gear is operated in a waist-deep or shallow portion of the water, where the line is lowered until the fish bites the hook. Once the fish bites the bait, the fisherman will pull the rod upward to get the catch.

The fishing method continues until today because the materials can be gathered easily, and the procedure is simple. Also, the method is sustainable since it only catches a few fish in accordance with their needs.

4. Fishing Using *Bidyo* (Improved Spear Gun)

The term *bidyo* can be interchangeably used as a gear (wounding gear) or a method (spearfishing). The gear consists of *kaha* (wooden frame resembling a gun), *guma* (rubber), *pisik-pisik/pisgaan* (push button made from the handle of a prong, either spoon or fork), and *hara/baslay* (arrow). The arrow is a long metal rod with barbed points on its head. Once it is attached to the gear, it is called *baslay*, which penetrates the prey's flesh. To operate this, the rubber sling is stretched once the *baslay* is loaded to provide propulsion to shoot. The *pisgaan* is pressed to disengage the arrow from the *sang-itan* (sear), allowing it to move forward rapidly.

The gear was created to answer the limitations of spearfishing. It has two kinds, which are *dipundo* and *dibuhi*. *Dipundo* is only used to catch small fish that thrive on the rocks. The term *dipundo* is derived from the word “pundo” which means “stop” in which the arrow does not detach from the gear once the *pisgaan* is pressed. On the other hand, *dibuhi* catches bigger fish and can reach a distance of 3 to 4 meters when the arrow is released. It is derived from the word “buhi” which means “release” since the arrow releases when the push button is pressed.

The fishing gear is still existing and used by the fishermen because of its versatility since it can adjust with the distance. Also, the *manuglente* (diver) used it as a goad when he did *pagsalom* (diving).

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5. Fishing Using *Pampalipong* or Ichthyotic Plants

Pampalipong has been used by many cultures around the globe as part of their adaptability to survive during hunting and gathering. It is a method of catching fish using ichthyotoxic plants gathered from the forest to stunning the fish. The plants serve as a natural poison that helps fishing become easier compared to using hands alone. This is done by pounding and squeezing the leaves or fruits using a stone to extract the juice.

Until today, the indigenous people of Tapaz and Dumarao employed this as part of their traditional fishing method. They used *tubli* (*Derris elliptica* Benth) and *tuba* (*Croton tiglium* Linn) plants by pounding the fruits to extract the juice and spread it in the shallow portion of the water to stun the fish. On the other hand, the Ati called this method “ginahiloan” or “poisoning” where they used *tig-ao* (*Callirpa cana* Linn), which they get from the forest of Mt. Tag-aw. However, the use of *pampalipong* in Jamindan, Capiz, is regulated by the Panay Bukidnon, especially on their sanctuary called *seperado* (protected land). It is a community-based approach to conservation by the indigenous people by assigning a sanctuary, either a river or a forest, that prohibits the excessive extraction of resources, which is attributed to a formal form of social control among the community members.

These indigenous environmental conservation practices protect the river, especially with *sapa* (stream), which they believe is a breeding ground for the fish with *bihud* (roe). Once the fish with roe is caught during fishing, they are obliged to release it, allowing them to reproduce. Excessive use of ichthyotoxic plants is regulated, and cleanliness is strictly observed. Violators of the implemented rules and regulations face consequences once proven. The punishment depends on the severity of the offense, by not allowing them to go fishing for a week, month, or year or even taking a bath within the protected areas.

6. Fishing Using *Taon*, *Binawgon* and *Pukol* (Fish Trap)

Panay Bukidnon used *taon*, *binawgon* and *pukol* as fish traps to catch sili (*Anguilliformes*), *urang* (*Macrobrachium rosenbergii* sp.), *unog* (family Gobiidae), and other fishes. *Taon* is made of bamboo woven into a cylindrical shape. The mouth is wide open, and the base is narrow. *Binawgon* is also similar to *taon*. It is made of bamboo woven like a trapezoidal cylinder. The base serves as an entrance for the fish and has a slat pointing towards inside the gear. Another device is *pukol*, constructed like *binawgon* with an entrance called *garaw* or *galaw* to prevent the fish from escaping once it enters the trap. It has uway (rattan) weaved on its body, allowing the device to adjust according to the size of the fish caught. The method is simple because the device's opening is placed facing against the water flow, allowing the fish to enter.

Nowadays, they use nylon as netting materials to enclose the bamboo frame of the gear so that it can last long. Though the indigenous people of Capiz changed the use of netting material, the structure of the gear remained the same. This shift is due to the modernization of fishing that influenced their traditional knowledge.

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7. Fishing through *Pagsalum* or Diving

Salum (dive) is a fishing method done underwater with or without gear to get food such as fish and other valuable marine and freshwater organisms like shells. It has been part of the hunting and gathering practices of the people near the bodies of water to explore other resources underneath.

In Panay Bukidnon, the person who does *pagsalum* (diving) is called *manuglente*. The term was taken from the word *lente*, which refers to a gear resembling goggles used to protect the eyes of the fisherman. The *lente* has a bamboo frame patterned after the eyeglasses; thus, it is called *antipara* (eyeglasses). A glass that they get from the lowland is attached to the frame using *tagok* (resins) of the *talolot* tree. A rubber sling is used to connect the improvised goggles to the head of the fisherman. As time passes by, the materials used for the *lente* also evolve. These indigenous people used wood as a new framing material because of its comfort and durability compared to *kawayan* (bamboo). At present, aside from wood, they are able to utilize other materials, such as Polyvinyl chloride (PVC).

Being a *manuglente* cannot favor anyone, and one must have *manamit it ginhawa* (strong respiratory endurance). Also, other gears are used, such as *sarapang*, *sagangat*, and *baslay*, during the fishing operation.

8. Fishing Using *Paniwtiw/tiwtiw* or Charm

Paniwtiw or *tiwtiw* is a fishing method employed by Panay Bukidnon long before to gather fish and crustaceans. They believe that using charms could attract fish. The entire process is more ritualistic since they use charms and prayers. Together with their spears, the fishermen go to the stream and perform the *paniwtiw* to make their fishing easier.

In Tapaz, Capiz, it is done first by looking for a small frog to be used for the rituals. The frog was encircled, and prayers were murmured. There is also *patibo-tibo*, where they look for a frog and catch it, just like *tiw-tiw*. A prayer is then uttered, and the frog is dipped into the water. After it is done, the *sili* (Anguilliformes) and *kagang* (Subphylum Crustacea) would come out once the ritual affected them.

On the other hand, in Jamindan, Capiz, *lago* (earthworms) is used as a medium for their rituals. This is done by tying the earthworm in the *paklang* (stalk) of the *payaw* (*Homalomena rubescens* Roxb.) plants. Then, it is dipped into the *lana sang lubi* (coconut oil) containing different woods and gathered and placed inside the bottle. It will then be dipped into the water, and fish will come out.

Currently, the method is no longer practiced in the community because they believe it has a harmful attribution. They believe that young boys use it as a charm to attract young women who bathe in the river. Like in the fish, a woman under the spell attracts the person who performs *tiw-tiw*.

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(13) Conclusions and Recommendations

Conclusions

The traditional knowledge systems of the indigenous peoples in the upland waters of Capiz reveal a dynamic and sustainable approach to fishing and environmental conservation. These traditional practices have been passed down through generations and remain relevant today. Some of these gears also evolve and undergo some modification to address the limitations and durability of the existing ones. As such, the Indigenous peoples learn how to improvise by hybridizing technologies and their knowledge systems (Ford et al., 2020). These traditional fishing gears and methods hold cultural significance for the indigenous Panay Bukidnon People and the Ati Community. Some notable insights from these methods encompass sustainable fishing methods designed to catch fish and other aquatic organisms while minimizing environmental harm and allowing fish populations to thrive. Resources conservation is also observed by releasing fish with roe to allow them to reproduce, which contributes to the preservation of the fish populations. Also, the use of ichthyotoxic plants (pampalipong) to stun fish is regulated, and cleanliness in the protected areas is strictly observed to maintain the health of the river ecosystem which is a community-based approach to conservation.

Recommendations

Based on the findings and conclusions of this study, several recommendations are proposed to support the preservation of traditional knowledge and the sustainable management of resources in the upland waters of Capiz:

Modernization of fishing gear influenced the traditional knowledge system of our indigenous people living in the upland due to acculturation. Thus, cultural preservation initiatives are needed to ensure the transmission of knowledge and practices to future generations.

Developing educational programs and materials to raise awareness about the cultural and environmental significance of traditional practices among local communities and visitors by organizing workshops and training sessions within the local communities. This is to educate residents, especially the younger generation, about their traditional fishing practices' historical, cultural, and ecological importance. Sessions can involve storytelling, hands-on demonstrations, and discussions led by community elders and knowledgeable individuals.

Furthermore, the local government unit (LGU) should formulate conservation policies that respect traditional practices to ensure their protection while achieving

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conservation goals that recognize the deep cultural and historical significance of these traditional practices and their role in the identity of the local communities.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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